In [2]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt

from skimage.feature import hog
from skimage import exposure
```

In [5]:
```
image = cv2.imread("butterfly.avif")

# Convert BGR to RGB for displaying with Matplotlib
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(8, 6))
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")
plt.show()
```



In [21]:
```
cv2.imwrite("original.jpg", image)
```

Out[21]: True

In [6]:
```
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

plt.figure(figsize=(8, 6))
plt.imshow(gray, cmap="gray")
plt.title("Grayscale Image")
plt.axis("off")
plt.show()
```



In [23]:
```
cv2.imwrite("grayscale.jpg", gray)
```

Out[23]: True

In [7]:
```
sift = cv2.SIFT_create()
```

In [8]:
```
keypoints, descriptors = sift.detectAndCompute(gray, None)

print("Number of keypoints:", len(keypoints))
print("Descriptor shape:", descriptors.shape)
```

```
Number of keypoints: 1026
Descriptor shape: (1026, 128)
```

In [9]:
```
sift_image = cv2.drawKeypoints(
    image_rgb,
    keypoints,
    None,
    flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS
)

plt.figure(figsize=(10, 7))
plt.imshow(sift_image)
plt.title("SIFT Keypoints")
plt.axis("off")
plt.show()
```

SIFT Keypoints



In [24]:
```
cv2.imwrite("siftImg.jpg",sift_image )
```

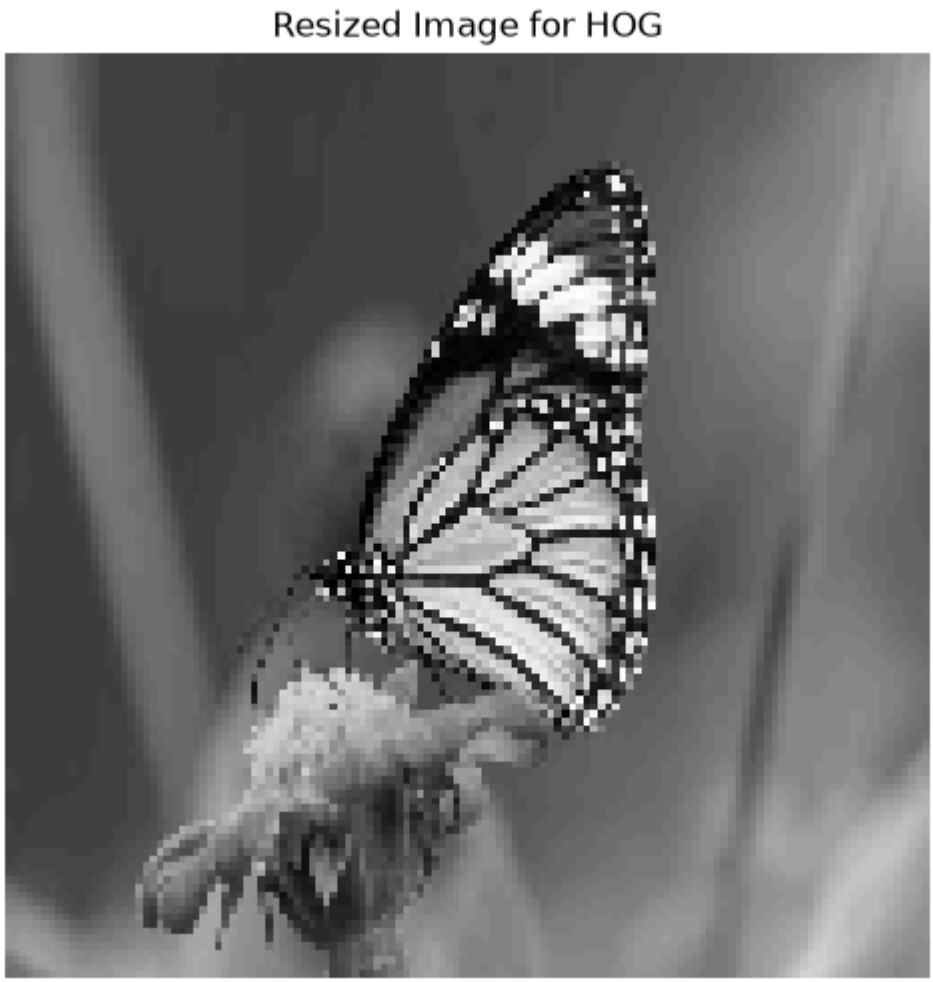
Out[24]: True

In [10]:
```
print("Number of keypoints:", len(keypoints))
print("Descriptor dimensions:", descriptors.shape[1])
```

```
Number of keypoints: 1026
Descriptor dimensions: 128
```

In [11]:
```
hog_image = cv2.resize(gray, (128, 128))

plt.figure(figsize=(6, 6))
plt.imshow(hog_image, cmap="gray")
plt.title("Resized Image for HOG")
plt.axis("off")
plt.show()
```



In [25]:
```
cv2.imwrite("hogImg.jpg", hog_image)
```

Out[25]: True

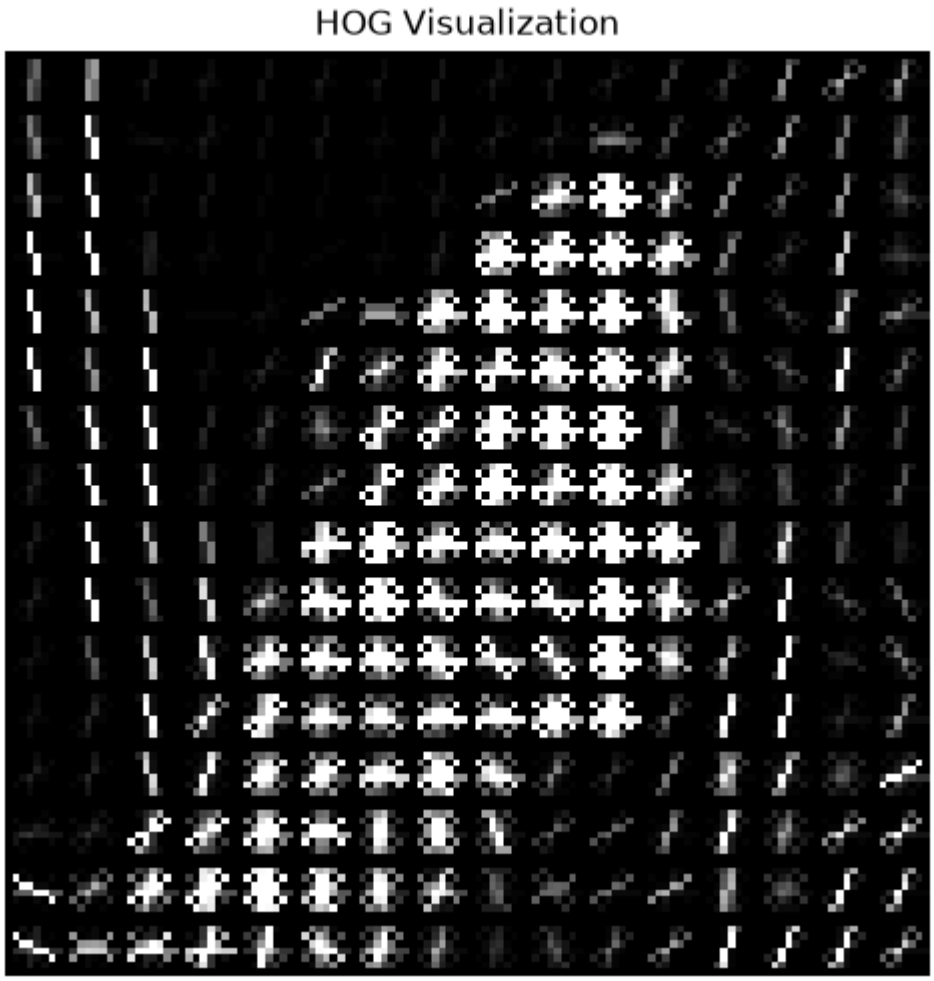
In [12]:
```
features, hog_visualization = hog(
    hog_image,
    orientations=9,
    pixels_per_cell=(8, 8),
    cells_per_block=(2, 2),
    visualize=True,
    block_norm='L2-Hys'
)
```

In [13]:
```
print("Number of HOG features:", len(features))
```

```
Number of HOG features: 8100
```

In [14]:
```
hog_visualization_rescaled = exposure.rescale_intensity(
    hog_visualization,
    in_range=(0, 10)
)

plt.figure(figsize=(8, 6))
plt.imshow(hog_visualization_rescaled, cmap="gray")
plt.title("HOG Visualization")
plt.axis("off")
plt.show()
```



In [27]:
```
cv2.imwrite("hogVisualization.jpg", hog_visualization_rescaled)
```

Out[27]: True

In [35]:
```
image2 = cv2.imread("butterfly2.jpg")
```

In [36]:
```
cv2.imwrite("similar.jpg", image2)
```

Out[36]: True

In [15]:
```
img1 = cv2.imread("butterfly.avif", cv2.IMREAD_GRAYSCALE)
img2 = cv2.imread("butterfly2.jpg", cv2.IMREAD_GRAYSCALE)

print(img1.shape)
print(img2.shape)
```

```
(491, 740)
(414, 738)
```

In [16]:
```
sift = cv2.SIFT_create()

kp1, des1 = sift.detectAndCompute(img1, None)
kp2, des2 = sift.detectAndCompute(img2, None)

print("Image 1 keypoints:", len(kp1))
print("Image 2 keypoints:", len(kp2))
```

```
Image 1 keypoints: 1026
Image 2 keypoints: 1497
```

In [17]:
```
bf = cv2.BFMatcher()

matches = bf.knnMatch(des1, des2, k=2)
```

In [18]:
```
good_matches = []

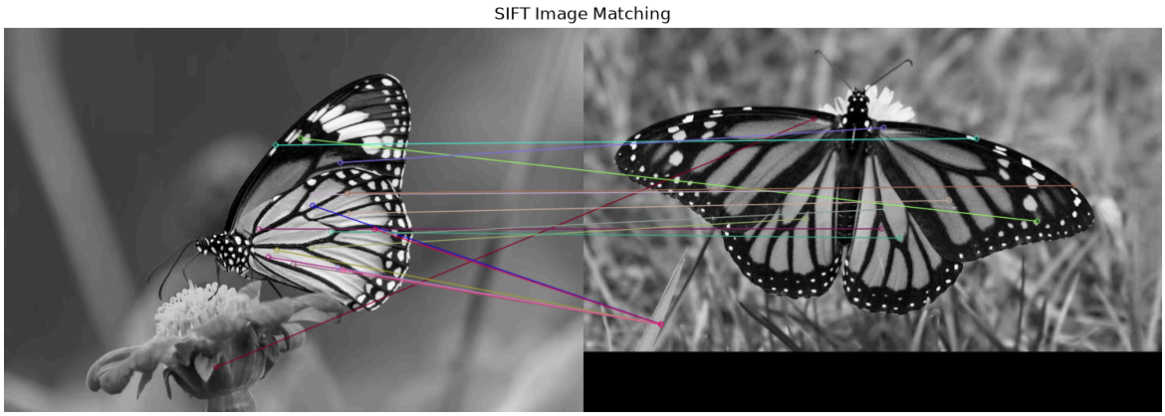
for m, n in matches:
    if m.distance < 0.75 * n.distance:
        good_matches.append(m)

print("Good matches:", len(good_matches))
```

```
Good matches: 18
```

In [19]:
```
matched_image = cv2.drawMatches(
    img1, kp1,
    img2, kp2,
    good_matches,
    None,
    flags=cv2.DrawMatchesFlags_NOT_DRAW_SINGLE_POINTS
)

plt.figure(figsize=(15, 8))
plt.imshow(matched_image, cmap="gray")
plt.title("SIFT Image Matching")
plt.axis("off")
plt.show()
```



In [28]:
```
cv2.imwrite("matchedImg.jpg", matched_image)
```

Out[28]: True

In [ ]: